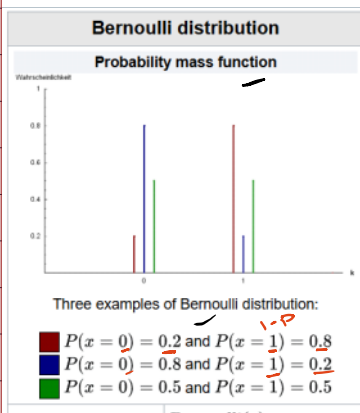




Definition:-

Bernoulli distribution is the discrete probability distribution of a random variable which takes the value 1 with probability P and the value 0 with probability $q=1-p$. Less formally, it can be thought of as a model for the set of possible outcomes of any single experiment that asks a yes-no question.

Eg: Tossing a coin $\{H, T\}$

$x \rightarrow$ is probability of outcome

$$P(x=1) = 0.5 \Rightarrow 50\%$$

$p \leq 1$

$$P(x=H) \rightarrow q \Rightarrow 1-p$$

$$q \Rightarrow 1-0.5$$

$$\Rightarrow 0.5$$

$p \Rightarrow$ Pass
 $q \Rightarrow$ Fail
 $q \Rightarrow 1-p$

P, q

(2) Student will pass or fail

$$P(x=Pass) = 0.4 \Rightarrow p$$

$$P(x=Fail) \Rightarrow q = 1-p$$

$$= 1-0.4$$

$$= 0.6$$

Parameter

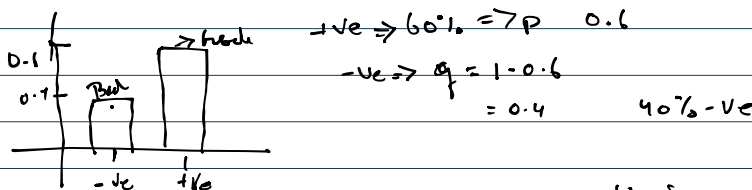
$$\therefore 0 \leq p \leq 1$$

$$q = 1-p$$

$$\Rightarrow K = \{0, 1\}$$

(Fail) success

Company products $\rightarrow +ve$
 $\rightarrow -ve$



$$K = \{1, 0\}$$

$$Pmf = p^k * (1-p)^{1-k}$$

$$q = k=1$$

$$P(x=k=1) = p^1 * (1-p)^{1-1} \Rightarrow p^1 * (-p) \Rightarrow p^1(1)$$

$$= p^1$$

$$P(x=k=0) = p^0 * (1-p)^{1-0}$$

$$q = 1-p$$

$$Pmf = \begin{cases} q = 1-p & \text{if } k=0 \\ p & \text{if } k=1 \end{cases}$$

$$p \quad \text{if } K=1$$

* Mean of Bernoulli distribution

$$K = \{0, 1\}$$

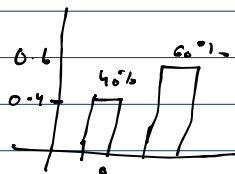
$$E(x) = \sum_{k=0}^1 k \cdot p(x)$$

$$= 0 \cdot 0.4 + 1 \cdot (0.6)$$

$$= 0 + 0.6$$

$$= 0.6 \Rightarrow p$$

$$p(\text{success}) \Rightarrow p$$



* Median of Bernoulli distribution

$$\text{Median} = \begin{cases} 0 & \text{if } p < 1/2 \\ [0, 1] & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$$

$$\text{Median} = \begin{cases} 0 & \text{if } p < q \\ 0.5 & \text{if } p = q \\ 1 & \text{if } p > q \end{cases}$$

* Mode

$$p > q \Rightarrow p \text{ is the mode} \\ \Leftrightarrow q > p \Rightarrow q \text{ is the mode}$$

* Variance

$$K = \{0, 1\}$$

$$\sigma^2 = x_i (0 - x_i) +$$

$$\sigma^2 = 0.4 (0 - 0.4)^2 + 0.6 (1 - 0.6)^2$$

$$= 0.4 \times 0.16 + 0.6 \times 0.16$$

$$= 0.16$$

//

$$\sigma^2 = q \cdot p \Rightarrow 0.4 \times 0.6$$

$$\sigma^2 = 0.24$$

$$\text{Standard Deviation} = \sigma = \sqrt{q \cdot p}$$

Ex. Bonus

$$\sigma = \sqrt{q \cdot p}$$